Strong vs Weak Acids

The strength of an acid refers to its ability to dissociate in solution.

Consider HCl:

$$HCl_{(aq)} + H_2O_{(l)} \longrightarrow Cl^-_{(aq)} + H_3O^+_{(aq)}$$

In this case, 100% of the HCl will dissociate in water. Therefore, the concentration of H_3O^+ will be equal to that of the HCl.

Some strong acids:

HCl	hydrochloric acid
HNO_3	nitric acid
H_2SO_4	sulfuric acid
HBr	hydrobromic acid
HI	hydroiodic acid
$HClO_4$	perchloric acid

In the case of acetic acid:

$$CH_3COOH_{(aq)} + H_2O_{(l)} \longrightarrow CH_3COO^-_{(aq)} + H_3O^+_{(aq)}$$

the dissociation of ions is not complete (actually, only about 0.5%!), therefore the concentration of H_3O^+ will not be proportional to the amount of CH_3COOH. This is considered a **weak** acid.

Other weak acids include HNO, HCN, and HF

In a similar fashion, there are strong and weak bases:

Some strong bases:

LiOH	lithium hydroxide
NaOH	sodium hydroxide
KOH	potassium hydroxide
$Ca(OH)_2$	calcium hydroxide
$Ba(OH)_2$	barium hydroxide

Some weak bases:

include all amines:
NH_3, CH_3NH_2, $(CH_3)_2NH$